FlexDCIM: A 400 MHz 249.1 TOPS/W 64 Kb Flexible Digital Compute-in-Memory SRAM Macro for CNN Acceleration

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***Abstract*— This work proposes a 64Kb fully reconfigurable SRAM compute-in-memory (CIM) macro for convolutional neural network (CNN) acceleration using a 65nm node. It supports operation up to 400 MHz. The fully digital operation of the proposed macro effectively removes the analog CIM design issues related to process variations, noise susceptibility, and data-conversion overhead. Hence, it offers no accuracy loss, high energy efficiency, and large area saving for computation. To support the digital computation, a new area-efficient Digital Processing Unit (DPU) is proposed which is equivalent to 8.75T per bit storage. Moreover, the proposed macro features full precision reconfigurability (1b to 8b) for both input and weight, and fully flexible input activation ranging from 1 to 64 parallel inputs. It makes the proposed macro feasible for different neural network topologies. Removing sense amplifiers (SAs) for the memory mode of the proposed design suggests additional area and power savings. The proposed CIM macro achieves an energy efficiency of 249.1TOPS/W and a throughput of 819.2 GOPS.**

***Index Terms*— Compute-in-memory, SRAM, digital-based CIM, CNN accelerator, MAC operation.**

I. Introduction

With the aggressive development of AI technology, specialized hardware is required for processing massive and parallel data. Convolutional neural networks (CNN) are predominantly involved in classification and recognition missions. However, traditional hardware employing the Von Neumann architecture struggles with high power consumption due to the frequent movement of data between memory and processing units. Compute-in-memory (CIM), as a promising solution, avoids this issue by breaking down the "memory wall" and allowing data processing to be carried out directly inside the memory [1], [2].

CIM designs can be classified into analog CIMs and digital CIMs depending on the signal type for multiply-and-accumulate (MAC) results. Analog CIMs are highlighted for their excellent throughput and energy efficiency [3], [4]. However, analog-based designs face challenges due to non-ideal performance from PVT variations, large area and energy overhead of ADC, and conversion precision loss. Alternatively, digital CIM has gained attention in recent years, providing more robust and scalable solutions [5], [6], [7], [8], [9], [10], [11], [12], [13], [14], [15]. Digital CIM employs digital logic to handle multiplication, offering easier architectural integration and greater tolerance for nonidealities.

To overcome the analog CIM issue, [7] introduced the dynamic-logic-based ADC-less SRAM CIM macro using signed-INT8 operations with reconfigurable bitwise operations. However, dynamic logic circuits can be less reliable and prone to variability, which may affect long-term stability and accuracy. CONV-SRAM [4] proposed an energy-efficient SRAM in-memory dot-product computation optimized for low-power CNNs and edge computing applications. However, it suffers from limited precision due to binary weights and doesn't meet the needs of modern AI applications requiring higher precision. Therefore, the reconfigurable CIM macros [5], [16] supporting 1-16b precision were proposed, those feature the column-MAC architecture with bit-serial computation. However, the bit-serial computation can increase latency, especially in high-precision scenarios, making the design slower for complex applications. Reference [17] uses 6T SRAM for in-memory computation but suffers from limited scalability and data flipping issues due to the common read/write port of 6T SRAM cell structure, which may struggle with high-precision operations. Reference [14] proposes a reconfigurable full-precision digital ReRAM-based CIM macro for AI applications, emphasizing bit-precision control. However, ReRAM-based designs can suffer from endurance issues, affecting long-term reliability in memory-intensive applications.

As a result, despite the recent work the digital CIM designs still face various design challenges. The multiplication and adder tree structures used for digital CIM further consume substantial area and energy, limiting the performance scalability. To address these challenges, recent research [6], [18] explored the use of approximate computation in digital CIM designs. However, the approximate arithmetic reduces accuracy limiting the architecture's use in precision-sensitive applications. Beyond approximations, researchers have also explored block and data-level

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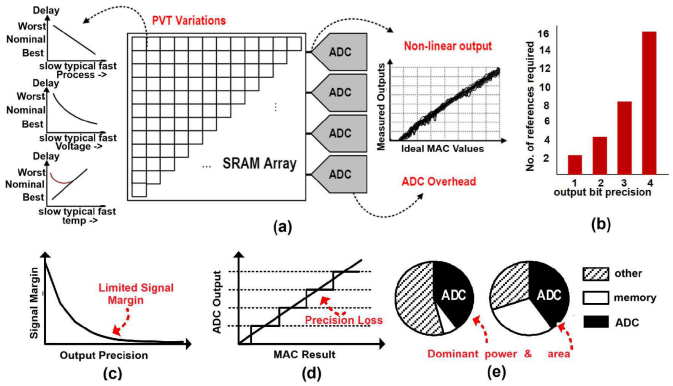


Fig. 1. Design challenges in analog-based CIM macros.

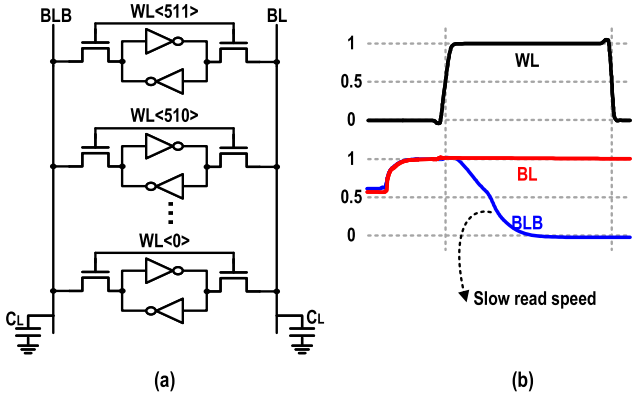


Fig. 2. (a) A column of 6T SRAM bit-cells, and (b) the read-out operation.

sparsity [8], [9], [13] to further reduce the computational load and enhance energy efficiency. Leveraging sparsity for energy efficiency enables CIM-friendly architectures, but sparsity introduces inefficiency in dense data processing and may reduce performance for non-sparse data. Also, it require specialized sparse training and additional hardware to handle the sparse data, which can add complexity to the overall system.

The other studies of [7], [16], and [19] proposed digital CIMs for no accuracy loss processing. Nevertheless, the reported CIM designs are restricted by the limited bit precision reconfigurability (reconfigurable to only 1, 4 or 8b) which limits their use for only fixed neural network topologies. Moreover, high-frequency operation of CIM macros is another requirement to support the complex CNN processing for handling the massive data.

Looking into the above said design challenges of CIM macro, this work proposes a new digital CIM architecture. Major contribution of the work is as follows:

1) Fully digital MAC operation with no accuracy loss.
2) Full flexibility to reconfigure for any input and weight precision from 1 to 8b.
3) Suggests a new sense amplifier (SA) less SRAM read scheme to reduce the SA area and power overhead during CIM mode.
4) A customized power-gated and clock-enabled full adder design to reduce power consumption.

The rest of this article is organized as follows: Section II describes the CIM design challenges solved in this work. Section III presents a new bitcell design for digital and reconfigurable CIM implementation, while section IV explains the overall architecture of the proposed CIM and its function. Section V highlights the mapping scheme of CIM macro for CNN applications, and section VI discusses about the performance analysis and measurement results. Finally, section VII concludes the work.

II. CIM Design Challenges Solved in This Work

A. Analog Non-Idealities

The analog-based CIM prototypes face intrinsic issues from the analog-intensive circuits which include the non-ideal performance due to PVT variations, ADC overhead in terms of area and power, and ADC conversion precision loss [11], [12], as can be seen in Fig. 1. Those challenges ultimately influence the accuracy of the CNNs. To overcome those issues, calibration techniques such as error correction algorithms, adaptive tuning etc are necessary but introduce additional design complexity. Large number of reference circuits required for multi-bit outputs of ADC put additional overhead for CIM implementation as shown in Fig. 1(b).

B. Bitline Parasitics Issue

Conventionally, the 6T SRAM cell based compuation is performed by discharging the pre-charged BL/BLB which depends on the weight value stored as the complementary data Q/QB of bitcell. However, the bitlines discharging speed is highly influenced by their load capacitance. This further degrades with the increasing column density of the bit-cell array. Fig. 2 illustrates an example of a memory array column consisting of 512 6T SRAM cells. As a result, the large load caused by the access devices of these 512 cells and the routing metal lines slows down the bitlines discharging. Since those bitlines voltages are evaluated to generate the computation data, the gradual discharging operation limits the read-out speed and thus increases the processing latency of the CIM operation. Hence, such bitcells with large column density don't fit for the high-frequency CIM operations.

C. Limited Reconfigurability

Digital CIM architectures often face the challenges of limited input and weight precision reconfigurability. Limited precision reconfigurability in CIM architectures restricts the ability of diverse workloads adaption, forcing designers to choose a "one-size-fits-all" approach that may not be optimal for all applications. Although, the input bit precision reconfiguration is enabled in the literature by serially applying the various bits of inputs. However, the weight precision reconfigurability is yet to be solved in the reported works [7], [16], [19].

D. Data Movement Complexity

Digital CIM designs need to balance computation and memory access. Frequent data loading is required when computations are performed in memory, leading to access bottlenecks, and reduced energy-efficiency [17] as can be observed

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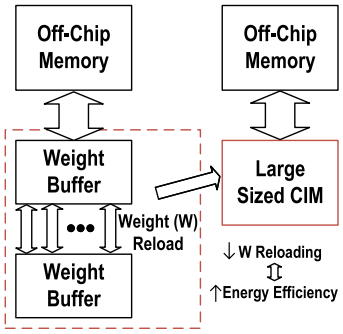


Fig. 3. Energy efficiency depends on weight reloading.

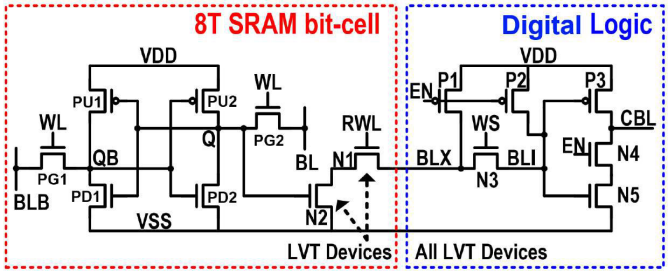


Fig. 4. Proposed digital bitcell for dot-product implementation.

TABLE I
CIM MULTIPLICATION

		Weight (Q)	
		0	1
Input (RWL)	0	0	0
	1	0	1
	No Input	0	0

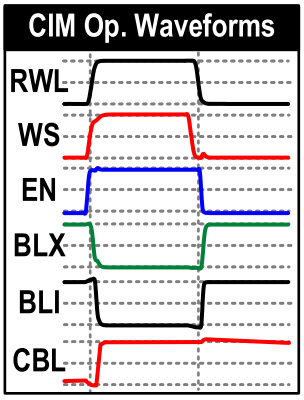


Fig. 5. Operation waveforms of proposed cell during computation for weight=‘1’.

TABLE II
BITCELL CONTROL SIGNALS IN DIFFERENT MODES

	Write 1/0	Read	Hold	Compute
WL	1	0	0	0
RWL	0	1	0	1
BL	1/0	Pre-charge		
BLB	0/1	Pre-charge		
WS	0	1	0	1

from Fig. 3. Various memory management strategies, such as interleaving, pipelining, and intelligent scheduling, are adopted to handle the memory conflicts and maximize effective memory bandwidth, however the limited energy-efficiency (TOPS/W) has still been a design issue.

To address all the above issues, this work proposes a fully-digital 64Kb SRAM CIM that is fully-reconfigurable from 1-8b of weight precision without any accuracy loss. Removal of ADC and sense amplifier from the architecture contributes to improve the area and energy-efficiency of this macro.

III. PROPOSED DIGITAL SRAM CIM IMPLEMENTATION

A. Proposed Bitcell Structure for Digital Computation

To support the high-frequency digital computation, a new bitcell structure is proposed. The proposed digital bitcell strucutre consists of an 8T SRAM bit-cell and a digital logic as shown in Fig. 4. The 8T SRAM cell part is formed of a 6T latch and two low-threshold voltage transistors (LVT) N1 & N2. The N1 & N1 transistors help to quickly discharge the precharged BLX node voltage of digital logic whenever RWL (input) and Q (weight) both are ‘1’. The digital logic part having six LVT transistors (P1-P3 & N3-N5) evaluates the BLX value and generates the final dot-product CBL at the output inverter formed of P3-N4-N5.

The truth table for the dot-product implementation of proposed cell is shown in Table I, while the operation waveforms with different control signals are shown in Fig. 5. Before each computation cycle, BLX and BLI are pre-charged to VDD by forcing EN at ‘0’. Once the EN and weight select signal (WS) go high, BLI node voltage is evaluated through the path N3-N1-N2 according to the respective RWL (input activation) and Q (weight) values. This evaluated BLI is reflected at the computation bit-line (CBL) as the digital dot-product. The use of N4 helps to preserve the CBL value until the next evaluation cycle. This helps to reduce the unexpected output glitch and dynamic leakage. Table II demonstrates the status of bitcell control signals for different operations.

B. Bitcell Area Optimization

Since a smaller memory can not store and process the huge data at the same time, and hence requires frequent energy-consuming weight reloading for computation [17], as shown in Fig. 3. Therefore, the energy-efficiency improvement can be supported with a larger sized memory array. The proposed digital cell incorporates this energy efficiency improvement using 8× increased weight storage than its computation capacity. This is achieved by sharing eight 8T SRAM part of digital bit-cells with a single digital logic as shown in Fig. 6. Therefore, the number of transistors overhead per bit storage is $6/8 = 0.75$T, and the equivalent number of transistors for each unit bit-cell is 8.75T which is hereafter referred as 8×1 Digital Processing Unit (DPU). It should be

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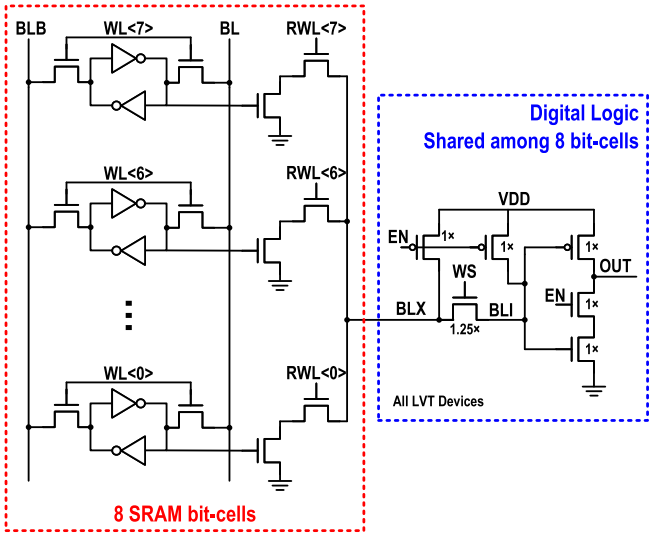


Fig. 6. Proposed 8 × 1 Digital processing unit (DPU).

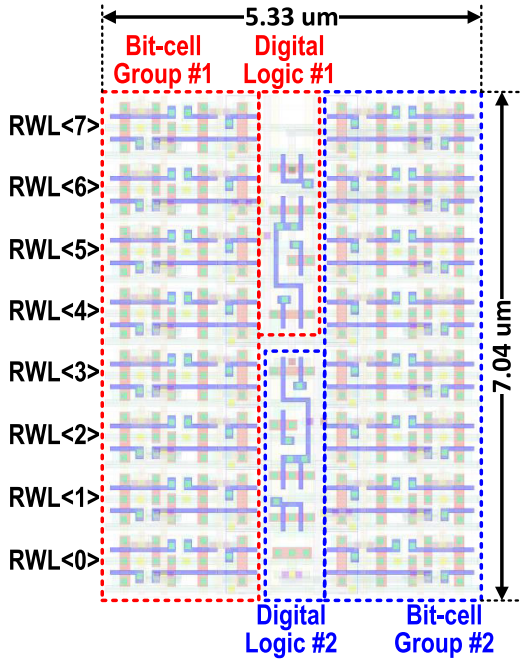


Fig. 7. Layout design for two adjacent 8 × 1 DPU.

noted that this DPU allows only a single RWL activation at a time while the other RWLs remain inactive based on the selection logic. Hence, compared with the 10T and 12T used in [4] and [19], the proposed DPU saves 12.5% and 27% transistors for 1b dot-product computation. For the layout drawing, two consecutive DPUs are flipped and coupled to reuse the area overhead caused by the digital logics of a single array column, as shown in Fig. 7.

C. Improved Performance for High-Frequency Computation

The proposed DPU doesn't face the large bitline parasitic issue due to the inherent divided bitline structure of proposed DPU. To generate the digital output, proposed DPU evaluates the BLI node voltage which depends on

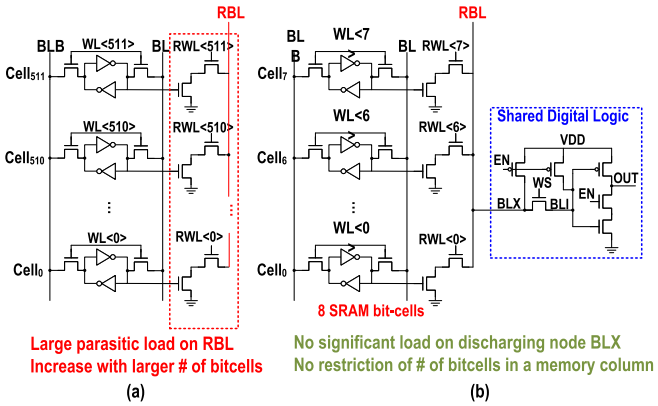


Fig. 8. (a) Conventional memory column during read: large parasitic load on RBL limiting the performance and signal margin and (b) inherent divided bitline feature of proposed bitcell removes such limitations.

the BLX discharging. Since, BLX is shared with only eight bit-cells irrespective of the array column density unlike the conventional SRAM array where multiple column access transistors remain connected to the single BL[1]. This ensures a rapid dot-product computation by a quick discharging of BLX for computation, please see Fig. 8.

Additionally, the larger LVT devices are employed in the computation path, which could provide a faster discharging speed for the read operation or dot product operation. Furthermore, the digital dot-product generation at CBL eliminates waiting for the actual input's arrival (RWL) to fasten the output generation. As soon as the precharge enable (EN) goes high to start the operation, CBL is instantly generated as '0' for the cases of input×weight= 0×0, 0×1, and 1×0. In case of input×weight= 1×1 also, the BLI/BLX discharge quickly due to the lower bitline loading and pulls up CBL at '1' by P3 LVT transistor of digital logic. In comparison, the NOR-based 10T digital implementation of [19] evaluates the dot-product only when the input is available, and results in the comparatively slower performance. Based on the bitcell simulation performed on 65nm node, it is identified that 10T [19] consumes 20ps of delay in contrast to the 9ps delay of the proposed bitcell. Moreover, the provision of selectively enable/disable the dot-product computation of proposed bitcell by the control signal WS further makes it superior than 10T [19] to provide full reconfigurability for the different weight precision.

IV. Proposed 64Kb Full-Precision Reconfigurable Digital CIM Architecture

A. Architecture of the Digital Processing Unit (DPU)

Fig. 9 demonstrates the proposed 8 × 8 digital module for multi-bit weight precision (8b W for this case), which consists of eight 8×1 DPUs sharing the same horizontol control signals (WL<7:0> and RWL<7:0>). Hence, n^{th} bit of an 8b weight is stored in an 8T SRAM part of nth DPU of this 8×8 module sharing the same input through RWL. EN<7:0> for eight individual DPUs are activated simultaneously to precharge BLX and BLI nodes. The WS<7:0> are activated according to the required weight (W) bit precision. Therefore, in one

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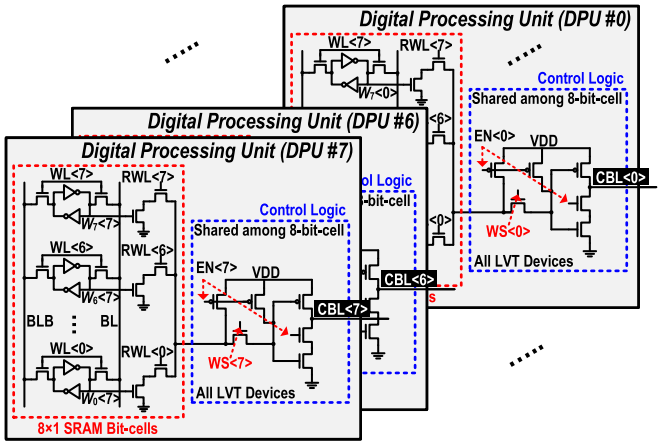


Fig. 9. 8×8 Digital sub-block (SBLK) for 8b weight implementation using eight 8×1 DPUs.

Decode Method
WS<0>=1
WS<1>=WSEL<2>+WSEL<1>+WSEL<0>
WS<2>=WSEL<2>+WSEL<1>
WS<3>=WSEL<2>+WSEL<1>WSEL<0>
WS<4>=WSEL<2>
WS<5>=WSEL<2>(WSEL<1>+WSEL<0>)
WS<6>=WSEL<2>WSEL<1>
WS<7>=WSEL<2>WSEL<1>WSEL<0>

Fig. 10. Decoding method.

CIM cycle, an 8×8 module performs 1b×8b multiplication for IN×W and generates 8b digital output (CBL<7:0>).

B. Reconfigurable Weight Precision From 1-Bit to 8-Bit

The neural network topology used in different AI edge devices is not fixed and varies for different applications. Therefore, the reconfigurability of input and weight precisions become essential to make it feasible for multiple applications. The input precision reconfigurability could be implemented by the bit-serial applied method. However, the reconfigurability of the weight precision is not well solved in the reported works [7], [16], [19]. The WS signals are used in the proposed structure to provide the full weight precision reconfigurability. In the proposed 8×8 digital module, all the BLX and BLI nodes are precharged by EN<7:0> to initiate the computation cycle. Then, the connection between the BLX and BLI is controlled by WS<7:0> for the 8b W so that the BLI nodes can be evaluated to generate CBL<7:0> for the given IN and W values. Therefore, the selective activation of WS<7:0> provides the full weight precision reconfigurability. These WS<7:0> signals are enabled by the external signal WSEL<2:0>.

WSEL<2:0> can be decoded to WS<7:0> to implement any weight precision from 1-8b using the logic shown in Fig. 10. For example, the 1b W precision is selected by externally applying WSEL<2:0>=“000”, which activates WS<7:0> as “00000001” allowing only LSB DPU of 8×8 digital logic to evaluate BLI for the computation. The BLI node voltage for other seven DPUs remain precharged due to WS<7:1> as “0000000” and CBL<7:1> remain unchanged at “0000000”. Similarly, the 8b W precision can be triggered with WSEL<2:0>=“111”, where the decoder logic brings out WS<7:0> as “11111111”. Thus all the eight DPUs in the SRAM module will participate in the CIM operation. The higher bit precision (more than 8b) can also be easily realized using proposed scheme by employing multiple CIM macro with additional shift-and-add accordingly. Thus, the proposed digital CIM macro can be used in an extensive range of neural networks with arbitrary input and weight precision.

C. Overall Architecture of the Proposed CIM Macro

Fig. 11 illustrates the overall architecture of proposed 64Kb full-precision digital memory macro, which consists of four main blocks for: 1) core CIM array including the 512×128 bit storage and sixteen binary adder trees, 2) CIM and SRAM drivers, 3) control & pre-charging circuits, and 4) sixteen accumulators. The proposed macro operates in basic SRAM mode whenever the weight loading and normal read operations are to be performed while the CIM mode is used to accelerate the high-accuracy MAC operation of CNNs. The proposed macro can perform 64×16 parallel vector multiplications for different 1b IN and 8b W values, and generate sixteen 14b neurons output simultaneously through sixteen blocks (BLKs). Each BLK consists of 64 8×8 digital modules (hereafter reffered as SBLKs) and one binary adder tree, as shown in Fig. 11. The proposed macro also avoids the limited I_{ON}/I_{OFF} current ratio and sensing margin related issues for the SRAM mode due to its inherent divided bitline structure of various BLKs.

D. SA-Less SRAM Read Operation

For analog CIM designs, sense amplifiers and ADCs are essential for SRAM read and CIM operations respectively. Those blocks consume a large area and significant portion of the overall power. To avoid such limitations, the proposed digital CIM macro removes the need of sense amplifiers and ADCs without influencing the functionality. During the read operation, all the BLX and BLI nodes are pre-charged, and then RWL of selected row is activated to turn on the BLI discharging path. Note that only one bit of WS<7:0> is enabled based on the required column selection for data reading. Depending on the data stored in the selected bit-cell, the reading result is obtained at CBL output of bitcell and will be passed to the adder tree.

During read ‘0’ operation, all the inactive RWLs of unselected rows don’t allow the BLX and BLI discharging which leads the corresponding CBL output of bitcells to be ‘0’. For the selected bit-cell of activated row, stored ‘0’ data at Q disconnects the discharging path of BLX and BLI resulting in the CBL output as ‘0’. Therefore, the adder tree accumulates all these CBL inputs and generates the 14b output as Y<13:0>=“00000000000000”. This 14b adder output is fed to an OR logic to produce the single bit read output, as shown in Fig. 12(a).

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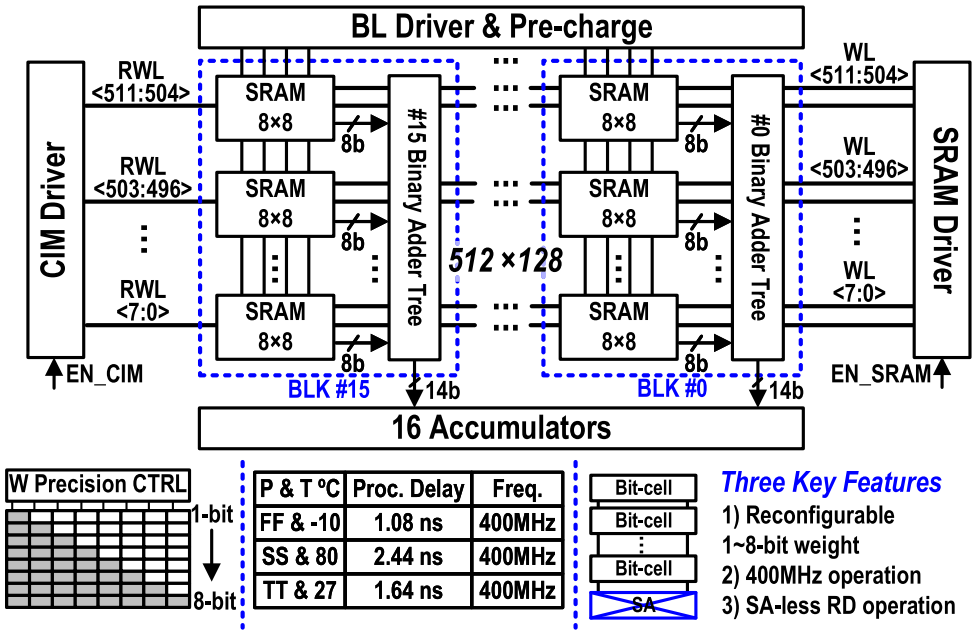


Fig. 11. Overall architecture of the proposed 64Kb digital CIM macro with three key features.

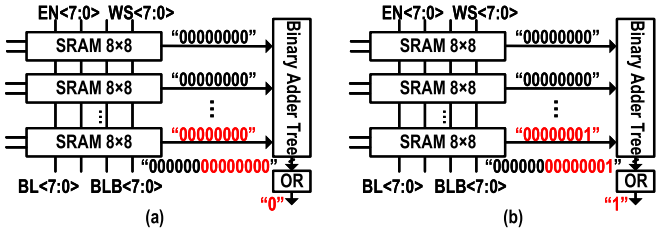


Fig. 12. (a) SA-less SRAM read "0" operation, and (b) SA-less SRAM read "1" operation (WS<0> = 1).

Similarly, for read '1' operation, the CBL output of selected bitcell is '1' while all other CBLs are '0'. The adder tree then accumulates the data and its output result Y<13:0> varies from "00000000000001" to "00000010000000" depending on the selected column through WS<n>. This is read as '1' by the OR logic, as shown in Fig. 12(b). It is important to note that unlike the CIM mode, only one bit of WS<7:0> is enabled '1' during SRAM read operation which allows only a single column activation out of each eight columns of every 8 × 8 SBLK digital module. The other seven bits of WS<7:0> remains inactive as '0'.

For the CIM operation, since the computation is conducted in the digital domain, there is no need to add the functional circuits for the domain conversion, such as the DAC and ADC parts. Moreover, the accumulation of the dot product can be obtained directly from the output of the adder trees in digital format.

E. Partial Sum Accumulation

Fig. 13 shows a single column of proposed macro which performs 64 MAC operations (using 64 8 × 8 SBLKs) for 1b IN and 8b W in a single CIM cycle. All those 64 digital MAC results are accumulated by the binary adder tree to generate a 14b partial sum for 1b input, as shown in Fig. 14(a).

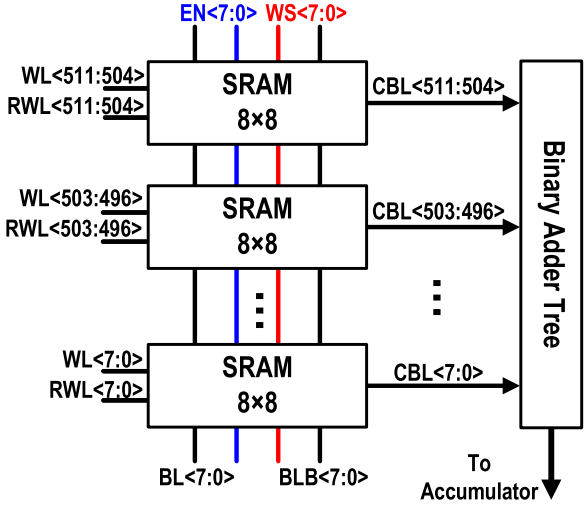


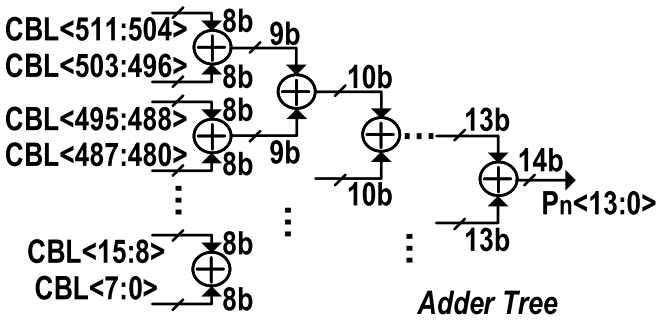
Fig. 13. One column of the proposed 64Kb CIM macro. Sixty-four 8b products of 1b input and 8b weight are generated at CBLs and passed to the adder tree for accumulation in one BLK.

Therefore, to implement the multi-bit input precision, bit-serial method is used where MSB and LSB of the binary weighted inputs are applied in first and last clock cycle respectively. Hence, n-bit input can be realized using n clock cycles. To combine all the partial sums, each 1b IN based 14b result is left-shifted in each cycle to maintain their positional weights and then accumulated with the partial sum values of the next clock cycles by an accumulator block. This shift and accumulation operation for 8-8b IN-W values generating a 22b digital output is illustrated in Fig. 14(b).

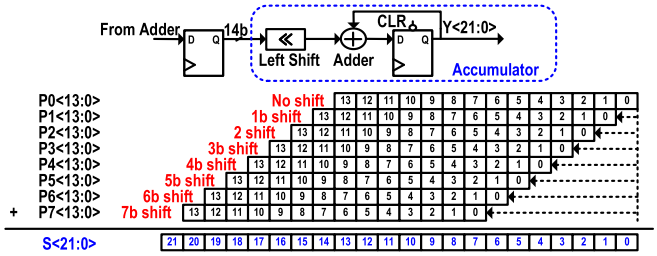
F. Power Gated Full Adder and Clock Enabled CIM Scheme for Power Reduction

The binary adder tree of the proposed digital CIM macro used a new customized power-gated and clock-enabled full

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(a) Accumulation in binary adder tree



(b) Post-CIM shift-and-accumulation

Fig. 14. (a) Binary adder tree to accumulate 8b product results from 64 SBLKs, and (b) post-accumulation method for one adder tree with 8b IN applied in eight CIM cycles.

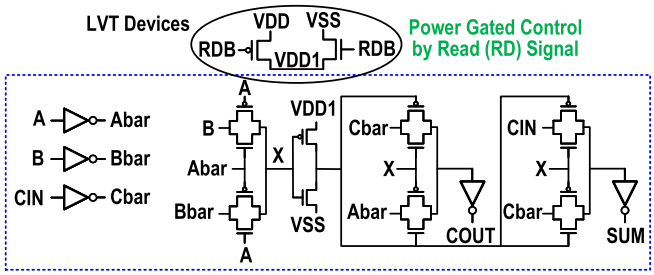


Fig. 15. Power gated full adder design.

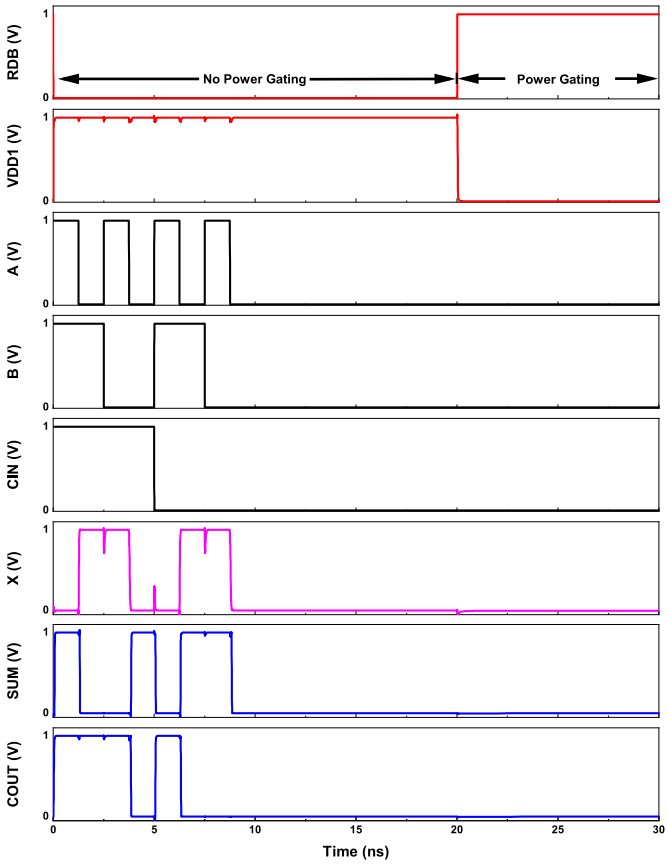


Fig. 16. Simulation waveform of proposed power gated full adder design.

adder (FA). This FA design aims to reduce the power overhead while maintaining the high performance and computational accuracy. As shown in Fig. 15, the FA internal supply voltage (V_{DD1}) is gated to either with V_{DD} or V_{SS} during read/compute or write/standby mode respectively. This power gating is controlled by a pair of P & N LVT transistors with their gate terminal enabled by inverted read signal (RDB). During idle or write mode when there is no computations or read operations required, power gating disconnects the complete power supply of the binary adder tree to significantly diminish the leakage current which could otherwise be a major contributor of the macro power. This feature is particularly effective in memory-intensive applications, where the adder tree may remain inactive for extended periods. Fig. 16 shows the simulation waveform of the proposed power-gated adder. The power gating mode of the proposed full adder helps to reduce its standby power by 80%.

The clock-enabled signal (EN) used in the digital control logic block of Fig. 4 and 6 complements the power saving by introducing the precise control over the adder's output transitions. Using EN, the CBL output coming through digital logic of a MAC unit is effectively prevented from any unintended switching or glitches between clock cycles. In conventional designs, these glitches or spurious transitions can occur due to propagation delays, causing temporary incorrect logic levels at the adder tree output. The clock-enabled scheme avoids such glitches by synchronizing the output changes with the clock. This significantly reduces the dynamic power consumption, as the unnecessary transitions are removed within the adder tree.

G. Multi-Bit Weight Implementation Analysis

An 8b W precision could be realized in the CIM macro either by the proposed scheme of using eight 8×1 DPUs as shown in Fig. 9, or the two accumulated modules of four 8×1 SRAM modules as suggested in [19]. The proposed scheme generates a digital partial sum for 1-8b IN-W by the adder tree with 8b adders at the first stage and 13b adders in the final stage, as shown in Fig. 14(a). On the contrary, another case of two 4b W according to [19] would require 4b and 9b adders at the first and final stage of its binary adder, respectively. Two of such macro columns would be needed to realize 8b W precision. However, the partial sum result from the higher four significant bits of the weight must be left-shifted by 4 to maintain its positional weight while accumulating the output of two 4b W. Therefore, the additional shift-and-add would introduce processing delay and thus decrease the

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TABLE III
COMPARISON BETWEEN TWO APPROACHES FOR 8-BIT WEIGHT IMPLEMENTATION

Two 4-bit weight			One 8-bit weight		
Type of Adder	# of Adder	# of FA	Type of Adder	# of Adder	# of FA
4-bit	64	256	8-bit	32	256
5-bit	32	160	9-bit	16	140
6-bit	16	96	10-bit	8	80
7-bit	8	56	11-bit	4	44
8-bit	4	32	12-bit	2	24
9-bit	2	18	13-bit	1	13

system latency. These delays become particularly problematic in systems where pipelining or parallel processing cannot fully mitigate them, as the shifted data must be properly aligned before accumulation can proceed. This sequential handling of shifting and alignment increases the overall critical path delay, especially for computations involving larger bit-widths. Hence, using a single module of eight 8×1 DPUs to implement 8b W is better than two modules of four 8×1 to enhance the latency.

Furthermore, two 4b W implementation as per [19] to realize 8b W consumes more hardware than the proposed scheme of using direct 8b W. The adder tree structure used in proposed 64Kb macro is shown in Fig. 14(a) which illustrates how sixty-four 8b dot product results are accumulated step by step through 32 8b adders at 1st stage to one 13b adder at final stage. According to a similar analysis done for the two 4b w implementation [19], a comparison table is shown Table III for the overall hardware requirement. It shows that the proposed 8b W implementation requires lesser number of full adders (FAs) at different stages of the adder tree when realizing the same bit of weight precision (8b). Therefore, the proposed scheme reduces the area overhead by lesser hardware requirement and improves the power consumption for high-precision computation.

V. MAPPING STRATEGY FOR CNN APPLICATIONS

The convolutional and fully connected layers can be accelerated with the CIM macro if the MAC operations are conducted in parallel [13]. Therefore, n weight filters of size W×W×C could be stored in n unit columns, where W is the height and width of the filter while C represents its depth. As shown in Fig. 17, one of the convolutional layers in the LeNet-5 is mapped to the CIM array. To perform the convolution operation, the input patch of similar size as that of weight filter ($4 \times 4 \times 4$) selected from the input feature map and applied to the CIM array. Then, $4 \times 4 \times 4$ vector multiplications of 1b input and 8b weight are obtained and accumulated in each BLK, and a total of 16 $4 \times 4 \times 4$ vector MAC operations are performed simultaneously in the proposed macro. Finally, with all the input patches applied to the array in multiple cycles, the convolution of the input feature maps and the filter weights are completed.

The mapping of a fully connected layer with the proposed macro is shown in Fig. 18, where the array utilization is higher than that for the convolutional layer. The CIM array

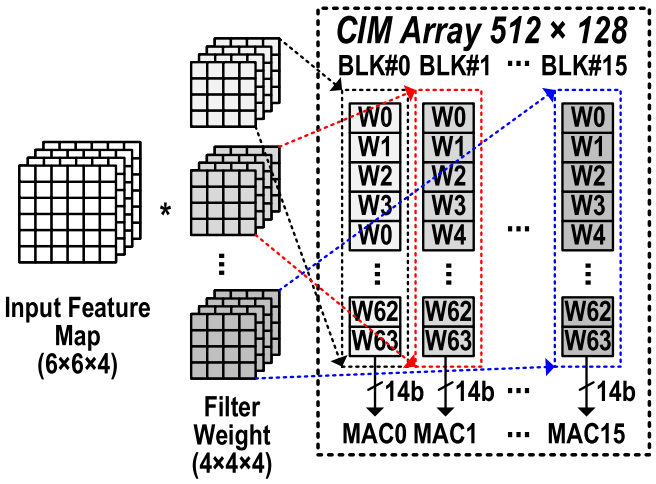


Fig. 17. Proposed 64Kb CIM macro mapping with the convolution layer; W_i in different BLKs represents 8b weight.

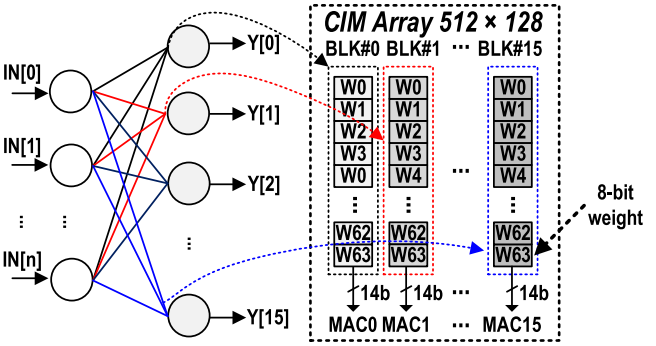


Fig. 18. Mapping of a fully connected layer with the proposed 64Kb macro.

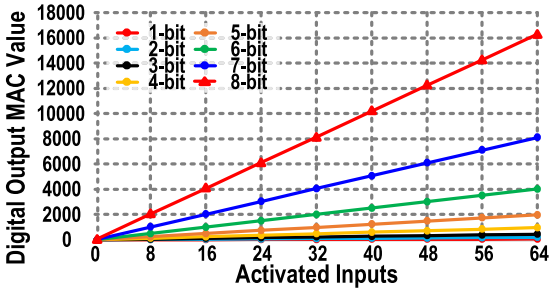
can be directly mapped with a fully connected layer of size up to 64×16. This can process sixty-four different data inputs simultaneously and generate sixteen parallel MAC results in each cycle. By activating a different single RWL<n> signal out of RWL<7:0> in eight different cycles, the 8× larger fully connected layer can be accelerated by the proposed macro without any weight reloading. Moreover, a larger size of 64×128 (8× larger) fully connected layer to generate 128 neuron outputs can be mapped with the proposed macro using eight computation cycles.

VI. PERFORMANCE ANALYSIS AND SILICON RESULTS

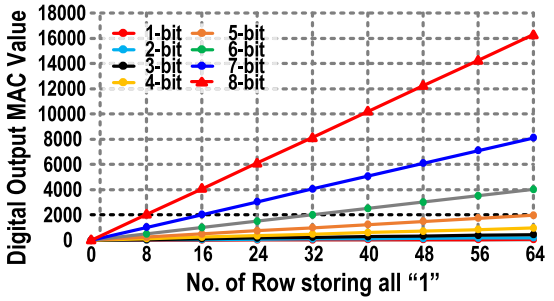
A. Linearity Analysis

The proposed 64Kb digital CIM is free from the linearity issues coming from the PVT variations, ADC offset, and the limited signal margin in analog-based CIM designs. This improved linearity of proposed CIM has been verified by two kind of simulation results. of the MAC values, two simulations with different activated inputs and the number of storing "1" rows are conducted, respectively. For the 1st case, all the weight values are '1' in the macro, and the number of activated inputs with value '1' are changed from 0 to 64. This is repeated for different 1-8b weight precision. As shown in Fig. 19(a), all the MAC values follow the straight line showing the ideal linearity for no accuracy loss. In another case, all the activated

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(a) MAC results for different no. of activated inputs with all weight values '1'



(b) MAC results with different no. of weight storing '1' while all the inputs are '1'

Fig. 19. Linearity analysis of CIM outputs.

TABLE IV
RECONFIGURABLE MAC VALUE RANGE FOR DIFFERENT WEIGHT PRECISION

Weight Precision	Highest MAC value	Output Precision
1-bit	00000001111111	7-bit
2-bit	00000011111111	8-bit
3-bit	00000111111111	9-bit
4-bit	00001111111111	10-bit
5-bit	00011111111111	11-bit
6-bit	00111111111111	12-bit
7-bit	01111111111111	13-bit
8-bit	11111111111111	14-bit

inputs are '1' and the number of rows storing all weight values as '1' are changed from 0 to 64, as shown in Fig. 19(b) which also illustrates the ideal linearity. Table IV illustrates the highest MAC values obtained for reconfigurable weight precision (1-8b) and its corresponding output bit precision when all the inputs are '1'.

B. Throughput Analysis

The proposed 64Kb macro implemented as 512 × 128b is fully reconfigurable for different precision. Therefore, its throughput has been analyzed for various bit precision and summarized in Table V. In case of 1-1b IN-W, there are 64 × 16.1024 dot products computed in a single cycle, where 64 and 16 are different IN and W values respectively. Since the maximum clock frequency is 400MHz, the throughput can easily

TABLE V
THROUGHPUT PERFORMANCE SUMMARY

Reconfigurability	1b∼8b Weight / 1b∼8b Input			
Bit Precision (W/IN)	1b/1b	1b/8b	8b/1b	8b/8b
Operation Cycles	1	8	1	8
# of Dot Product @ 1 cycle	64×16×1	64×16×1	64×16×8	64×16×8
Frequency (MHz)	400	400	400	400
Throughput (GOPS)	819.2	102.4	819.2	102.4

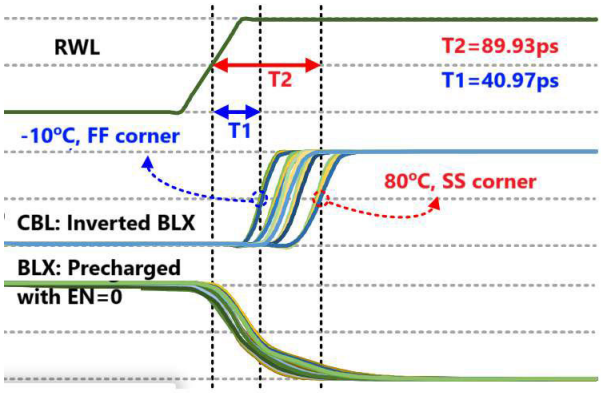


Fig. 20. Computation delay in different process and temperature corners.

be calculated in terms of Giga Operation per Second (GOPS) as 819.2 GOPS. In similar manner, 8-1b IN-W based throughput is estimated as 102.4 GOPS and the complete operation for it takes 8 CIM cycles. For the weight precision fixed at 8b and input precision being 1b and 8b, the total 8192 number of the dot products are performed in both the cases by the proposed bit-cell array using 1 and 8 cycles respectively. The corresponding throughputs for those two case are estimated as 819.2 GOPS and 102.4 GOPS respectively.

C. Process and Temperature Variation Analysis for High-Frequency CIM Operation

The basic operation of the proposed digital CIM macro depends on the precharging of BLX and its evaluation when the corresponding RWL is activated, as described in Section III A. Accordingly, the inverted BLX represents the output as CBL per dot product. Therefore, to confirm the process and temperature variation tolerance of the proposed design, the proposed 8 × 1 DPU was simulated for different process corners (TT, FF, FS, SF, SS) and temperatures (−10°C, 27°C, 80°C) values. Fig. 20 illustrates the simulation waveforms for the worst case of CBL generation when both the input and weight are '1'. It can be seen from the figure that the best and worst case delays (RWL -> CBL) are 40.97ps and 89.93ps for 1V supply, and FF/−10°C and SS/80°C corners respectively. Nevertheless, in all the cases the delay is less than 90ps to support the fast computation of the proposed CIM macro. The overall computation delay estimation for the proposed CIM is shown in Fig. 21. This delay is the sum of CLK -> RWL (T1), and RWL -> adder tree output Y<13:0> (T2) delays shown in the figure.

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TABLE VI
COMPREHENSIVE COMPARISON WITH STATE-OF-THE-ART WORKS

	ESSCIR'19 [16]	ISSCC'21 [19]	ISSCC'22 [7]	JSSCC'24 [18]	**This Work**
Technology	65nm	22nm	28nm	28nm	65nm
Supply Voltage	0.6-0.8 V	0.72 V	0.8 V	0.54-0.9 V	1 V
Array Size	16 Kb	64 Kb	32 Kb	16 Kb	64 Kb
Bit-cell	6T	6T	6T	8T	8T
***Bit-cell Area** (μm^2)	10.53	55.85	1.384	1.293	2.3452
***Macro Area** (mm^2)	0.2272	1.763	0.1616	0.1509	0.3136
CIM Weight Density (Kb/mm^2)	70.4	36.3	198	106	204
Input Precision	1-16b	1-8b	1-8b	2-8b	1-8b
Weight Precision	1-16b	4/8/12/16b	1/4/8b	2-8b	1-8b
Output Precision	8-23b	16/24b	21b	16b	14b
Energy Efficiency ($TOPS/W$)	117.3 (1/1b)	24.7 (8/8b), 89 (4/4b)	27.38 (8/8b)	102 (8/8b)	249.1 (1/1b), 110.7 (1/8b), 13.84 (8/8b)
Frequency (IN/W)	138 MHz	100 MHz (4/4b), 56MHz (8/8b)	42 MHz	20 MHz	400MHz (1/8b), 50MHz (8/8b)

*Scaled w.r.t 65nm technology

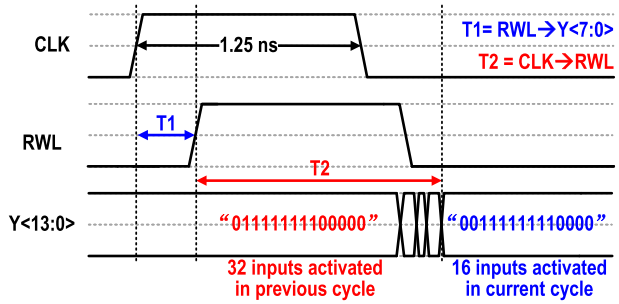


Fig. 21. Signal delay illustration in one CIM cycle; all stored weights are '1' for this case.

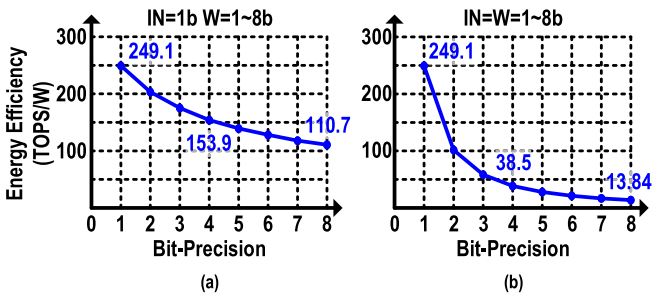
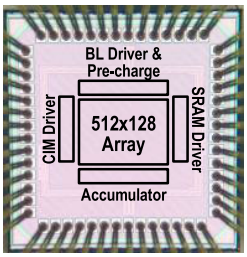


Fig. 23. Measured energy efficiency for different input and weight bit precision.



Process	TSMC 65nm
Supply Voltage	1V
Frequency	400MHz
Bit-cell	8T SRAM
Bit-cell Area	2.3452um²
Array Size	64Kb
Array Area	0.3136mm²

Fig. 22. Test-chip die photo and chip summary.

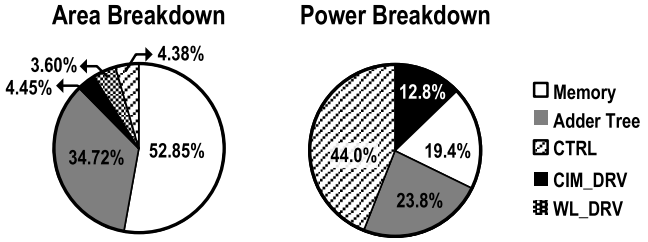


Fig. 24. Area and power breakdown.

D. Chip Micrograph and Summary

The proposed digital CIM macro has been fabricated using a standard 65nm technology node. The die micrograph and its performance summary are shown in Fig. 22. The measured energy efficiency with different IN and W bits precision is shown in Fig. 23. The proposed CIM macro achieves the energy efficiency of 249.1 TOPS/W for 1-1b IN-W bit precision while it provides 110.7 and 13.84 TOPS/W energy efficiency for 1-8b and 8-8b IN-W bit precision respectively. Fig. 24 illustrates the area and power breakdown details of the macro. It can be seen from the figure that memory bitcell array occupies 52.85% of the overall macro area while the adder tree area occupancy is 34.72%. The macro's control sub-block (including the MUX, BL driver, precharge and EN control circuitry) consumes a 44% of the overall power for 1-1b of IN-W bit precision. For CIM mode, the CIM Driver module controls the wordlines. Hence the power consumption of the WL driver has been considered under the CIM driver block. The comprehensive performance comparison of the proposed architecture with state-of-the-art works is summarized in Table VI. From the table, it can be concluded that the highest operation frequency, maximum weight precision reconfigurability, and no accuracy loss make this digital CIM macro attractive for the various CNN accelerators.

VII. CONCLUSION

This work has addressed the limitations of analog-based CIM architecture such as process variation, limited signal margin, poor-linearity, low-accuracy, and high area/power

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overhead due to data conversion. To solve such limitations a fully-digital area efficient 64Kb CIM macro has been proposed. Since, most of the existing digital CIMs face the issues of low-frequency, and large area occupied by the bitcell's digital logic and adder tree, this work solves those challenges also. To improve the area-efficiency and speed of the proposed digital CIM macro, a new digital bitcell has been proposed which shares a digital logic with eight 1b weight storage cells that results into a compact 8.75T equivalent bitcell. As a result, the overall CIM array weight density (Kb/mm^2) of proposed macro is 1.03×, 2.9×, 1.92× and 5.6× when compared with [7], [16], [18], and [19] for same technology nodes. Moreover, this work offers the full flexibility to reconfigure the CIM macro for any bit precision from 1-8b (or 1-16b) using one (or two parallel) macro supporting the diverse CNN applications unlike the other existing digital CIMs ([7], [18], [19]) those have limited degree of bit precision reconfiguration. Also, the proposed digital CIM macro supports upto 400MHz frequency of operation, at 65nm node, claiming it a high frequency candidate of digital CIM architectures. Furthermore, a new customized power-gated adder design for adder tree of this digital CIM could reduce the overall dynamic power by 38% during power gating mode. Therefore, the high-speed, area-efficient and low power operation of the proposed SRAM-based digital CIM macro claims to be an attractive choice for CNN processing in modern edge AI applications.

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